Unexpected Recommender System for Improving User Satisfaction

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Motivation

Classical recommender systems face the problem of

- **User Boredom** (repeated or similar recommendations)
- **Filter Bubble** (limited contents of recommendations)

Unexpectedness, on the other hand, is capable of

- **Overcoming the Overspecialization Problem** [Iaquinta 2010]
- **Broadening User Preferences** [Zheng 2015]
- **Increasing User Satisfaction** [McNee 2006]

Definition

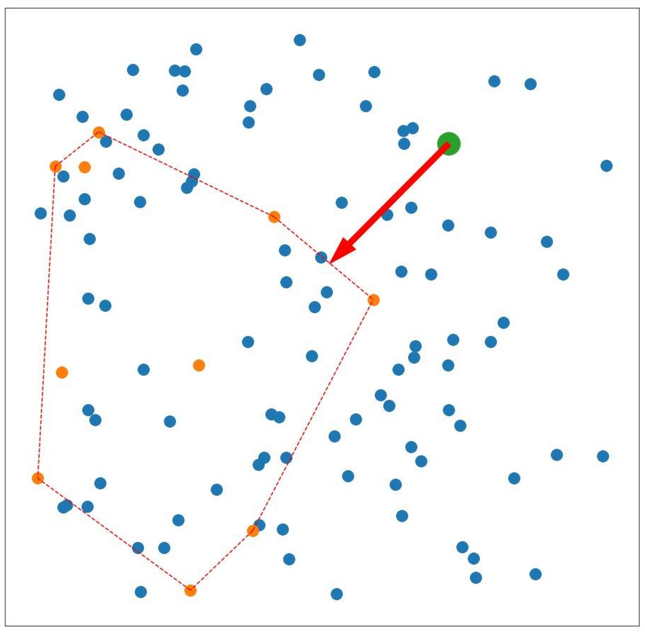
Unexpectedness is defined as the distance between the new recommendations and the user expectations. [Adamopoulos2015]

Goal

Unexpected Recommender System aims at providing novel and satisfying recommendations simultaneously to escape filter bubbles. [Li2019]

Overview of Completed Work

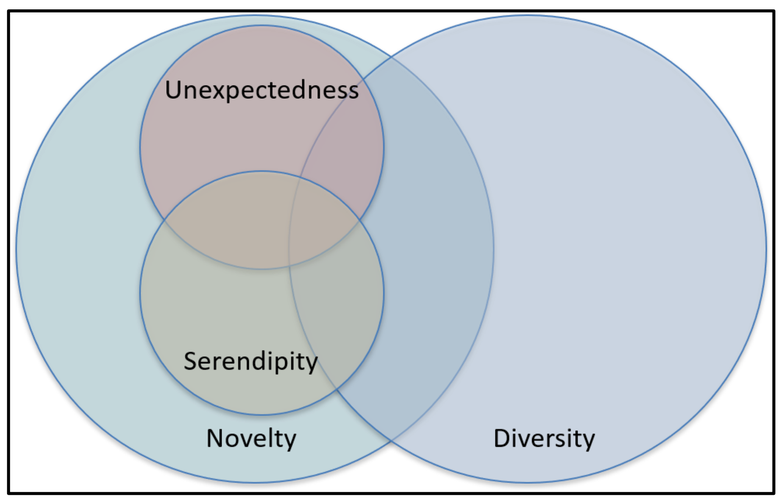
1. **Latent Modeling of Unexpectedness [RecSys19] [TIST] (Under Review)**



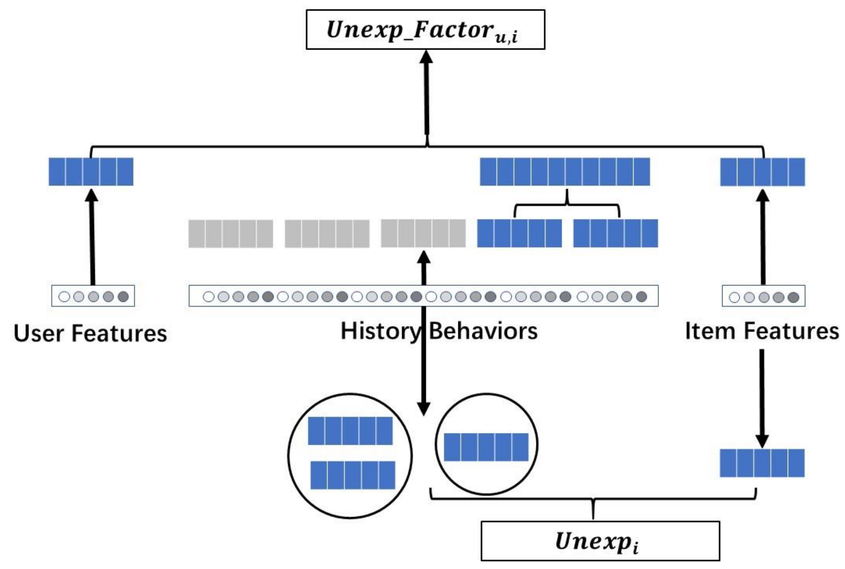
$$Unexp_{u,i} = d(i; LC(N_i))$$

$$Utility_{u,i} = (1-\alpha) * EstRating_{u,i} + \alpha * Unexp_{u,i}$$

2. **Survey on Unexpected Recommendations [TKDE] (Under Review)**



3. Session-based Personalized Unexpected Recommendation (Under Review at a conference)



$$Utility_{u,i} = b_u + b_i + r_{u,i} + f(unexp_{u,i}) * unexp_factor_{u,i}$$

Work in Progress

1. Reinforcement Learning for Unexpected Recommendations (In Preparation for RecSys20)

- Update unexpected recommendation policy in real-time based on current user feedback

2. Heterogenous Treatment Effect of Unexpected Recommendations on Consumers (In Preparation for Management Information System Quarterly)

- Measure the heterogeneous casual effect that the unexpected recommendation impose on different users

3. Variety-Seeking Behavior in Recommendation Scenario (In Preparation for Marketing Science)

- Identify the matching between variety-seeking in marketing literature and unexpected recommendations in practice

Challenges

1. How to identify user purposes at each session?

- Does the user want to continue watching this TV series? Or prefer to try something new?
- What would trigger the switch between different user proposes?

2. How to measure the long-term impact of unexpected recommendations?

- Multiple rounds of unexpected recommendations should result in broadening user interests.

3. How to identify whether unexpected recommendations could escape the filter bubble or not?

- How to quantify the filter bubble?

4. How to obtain better more precise estimation of user expectation?

- Can we combine empirical user study with statistical modeling to understand user expectation better?